

Chapter 12

Adverse Selection and Moral Hazard

"In the market for lemons, it is the sellers who know and the buyers who don't. The result is that the good cars are driven out by the bad."

— George Akerlof, *The Market for Lemons*, 1970

Before You Begin: Quick Self-Assessment

Answer each question before reading. Check your answers at the end of the chapter.

1. **Expected value:** A coin flip pays \$0 (tails) or \$100 (heads). What is the expected payoff? *Answer: \$50.*
2. **First-order condition:** Maximise $U(e) = 10e - 2e^2$ over e . What is the optimal e^* ? *Answer: $e^* = 2.5$.*
3. **Market failure:** Name two types of market failure studied in Week 11. *Answer: Public goods and externalities.*
4. **Asymmetric information (intuition):** You are buying a used laptop on Craigslist. The seller knows whether it has a hidden defect; you do not. Before you read the chapter: why might this cause a problem beyond just "unfair" pricing? *Answer: You cannot offer the right price for quality you cannot verify; high-quality sellers may not find your offer acceptable.*
5. **Risk aversion:** Contract A pays \$100 for certain. Contract B pays \$200 with probability $\frac{1}{2}$ and \$0 with probability $\frac{1}{2}$. A risk-averse person prefers _____. *Answer: Contract A (the certain outcome), even though both have expected value \$100.*

Connecting Week 11 to Week 12: A Third Type of Market Failure

Weeks 1–11 Established

Markets fail when prices don't capture all social costs (externalities) or when goods are non-excludable (public goods). In both cases, *prices exist* but reflect the wrong information.

Left open: What if one side of the market simply *cannot observe* what the other side knows or does?

Week 12 Analyses It

Adverse selection (hidden type): The seller of a used car knows its quality; the buyer does not. Good cars are crowded out.

Moral hazard (hidden action): After buying insurance, the driver knows how carefully she drives; the insurer does not. Effort falls below the social optimum.

Both problems arise from *information asymmetry* and require informational — not price — remedies.

*Key distinction: adverse selection is **ex ante** (before contracting); moral hazard is **ex post** (after contracting). This timing difference determines which tools fix each problem.*

Questions This Chapter Will Answer

- Why does George Akerlof's simple model of used-car markets win a Nobel Prize? What does it show that wasn't obvious before?
- When does a used-car market have only lemons left? When do peaches survive? What is the knife-edge fraction p^* ?
- How does education as a signal work — even when it teaches you nothing useful?
- Why do flat salaries produce zero CEO effort, while strong profit-sharing produces first-best effort?
- What is the incentive–insurance trade-off, and why does $\beta^* = 1/(1 + r\sigma^2)$?
- How did the 2008 financial crisis combine both adverse selection and moral hazard?

Chapter Road Map. This chapter analyses information as a source of market failure. We open (§12.1) by contrasting adverse selection and moral hazard and placing both in the broader landscape of market failures. We then study adverse selection in depth (§12.2): the Akerlof lemons model, the pooling/separating decision rule, the market unravelling cascade, and the three solutions (signalling, screening, certification). The Spence signalling model is developed with full mathematics and a diagram. Section 12.3 presents real-world applications — health insurance, credit rationing, and labour markets. Section 12.4 develops the principal–agent model of moral hazard from scratch: three compensation schemes, the derivation of optimal β^* , and the incentive–insurance trade-off. Section 12.5 compares adverse selection and moral hazard head-to-head. Three solved examples, a two-page summary, and twenty-five practice questions complete the chapter.

Learning Objectives and Topics

Each objective states what you will be able to do.

LO1 Define adverse selection; derive the pooling/separating condition; find p^* .

§12.2 Adverse Selection

LO2 Trace the unravelling argument through multiple rounds of exit; compute welfare loss.

§12.2 Unravelling

LO3 Explain signalling and screening; apply the single-crossing condition; interpret the Spence model ICC range.

§12.2 Solutions; §12.3

LO4 Set up the principal-agent model; derive optimal CEO effort under a linear contract.

§12.4 Moral Hazard

LO5 Apply the PC and ICC; solve for optimal β^* ; interpret the incentive-insurance trade-off with comparative statics.

§12.4 Moral Hazard

LO6 Distinguish adverse selection from moral hazard by timing and type of hidden variable; apply the lie-detector/camera test.

§12.5 Comparison

Notation Used in This Chapter.

p	fraction of peaches in the market	p^*	critical (knife-edge) fraction
$\mathbb{E}[\text{WTP}]$	expected buyer willingness to pay	$\pi(e)$	firm profit as function of effort
e	agent's effort level	$c(e)$	cost of effort ($c' > 0$, $c'' > 0$)
β	performance weight in linear contract	β^*	optimal performance weight
$\bar{U}$	agent's reservation (outside-option) utility	r	coefficient of absolute risk aversion
σ^2	variance of profit shock	ICC	incentive compatibility constraint
PC	participation constraint	FB	first-best (full-information) benchmark

12.1. Introduction: Why Information Matters

The competitive model assumes symmetric information. What happens when one side knows something the other doesn't — and why does this destroy markets?

Two Canonical Information Failures

Problem	Hidden variable	Timing	Direction of distortion	Tools
Adverse selection	Type / quality (fixed)	<i>Ex ante</i>	High-quality crowded out	Signalling, screening
Moral hazard	Action / effort (chosen)	<i>Ex post</i>	Sub-optimal effort	Performance pay, monitoring

Both arise from information asymmetry; both require informational — not price — remedies. The timing difference is the key diagnostic.

Case Study: The 2008 Financial Crisis — Both Failures at Once.

The 2008 crisis is the defining real-world illustration of adverse selection and moral hazard operating simultaneously and reinforcing each other. Understanding it requires keeping the two mechanisms separate.

Background. In the early 2000s, mortgage originators (banks and non-bank lenders) extended trillions of dollars in home loans. Rather than holding these loans on their balance sheets, they bundled them into *mortgage-backed securities* (MBS) and sold them to investors worldwide — a process called securitisation. This “originate-to-distribute” model created the conditions for both information problems.

1. Adverse selection (hidden type, ex ante). Mortgage originators knew far more about individual loan quality than the investors who purchased MBS. A local loan officer could see that an applicant had misrepresented income or had a spotty payment history; an investor in Frankfurt buying a tranche of 5,000 pooled mortgages could not. Investors, unable to distinguish safe from toxic loans, priced all MBS at an average. Once defaults mounted, investors realised that the pool was disproportionately toxic — originators had selected which loans to bundle and sell, retaining the better ones (or simply not caring, given the incentive structure

below). This is a textbook Akerlof-style unravelling: informed sellers (originators) and uninformed buyers (MBS investors), with bad assets crowding out good ones in the secondary market.

2. Moral hazard (hidden action, ex post). Once a bank originated a loan and immediately sold it into a security, it no longer bore the default risk. The bank's screening effort — verifying borrower income, appraising collateral carefully, checking employment history — was unobservable to MBS buyers. Knowing the risk would be passed on, banks had sharply weaker incentives to screen. The result: documented cases of "NINJA" loans (No Income, No Job, No Assets) approved and securitised within days. This is moral hazard: the agent (bank) chose hidden effort (poor screening) after the contract (securitisation agreement), and the principal (investors) bore the consequences.

3. Compounding factor: Rating agency conflicts. Credit rating agencies (Moody's, S&P, Fitch) were paid by the very issuers whose securities they rated. This created an incentive to inflate ratings — both an adverse selection problem (agencies knew more about model assumptions than investors did) and a moral hazard problem (agencies chose unverifiable rating rigour after the rating contract was signed).

Timeline of the two mechanisms:

Stage	Year	Mechanism	What happened
Origination	2002–06	Moral hazard	Banks under-screen; toxic loans created
Securitisation	2002–06	Adverse selection	Toxic loans bundled & sold to uninformed
Rating	2002–06	Both	Inflated AAA ratings obscure true quality
Unravelling	2007–08	Adverse selection	MBS market collapses as true quality revealed
Bailout	2008–09	Moral hazard	"Too big to fail" validates future risk-taking

Policy response (Dodd-Frank Act, 2010).

- *Skin-in-the-game rule*: originators must retain at least 5% of securitised loans — a direct response to moral hazard, making the agent a partial residual claimant.
- *Risk-retention requirements*: forced alignment of originator and investor incentives.
- *Enhanced disclosure*: reduces the informational gap (adverse selection) by requiring more granular loan-level reporting.

The crisis was not simply greed or fraud — it was a textbook case of both information failures operating at scale, with policy reforms designed using exactly the tools this chapter develops.

We begin with adverse selection — the hidden-type problem that can destroy entire markets.

12.2. Adverse Selection: Hidden Information

A seller knows whether her car is a lemon or a peach. The buyer doesn't. How does this single informational gap destroy the market for high-quality cars?

Adverse Selection (Hidden Information Problem): One party to a potential transaction has private information about a relevant characteristic — quality, risk, or type — that the other party cannot observe *before* the transaction.

Because uninformed buyers cannot distinguish high-quality from low-quality goods, they pay only the *average* expected value. This average price drives high-quality sellers out, leaving a disproportionate share of low-quality goods: the word “adverse.”

Timing: *ex ante* (pre-contractual). The information asymmetry exists before the deal is struck.

Adverse selection is ex ante. Moral hazard is ex post. This timing distinction is the most important diagnostic in information economics.

12.2.1. Akerlof’s Lemons Model: Setup

Two types of used cars:

Type	Seller’s reservation price	Buyer’s WTP	Gains from trade
Lemon	\$1,000	\$1,200	\$200
Peach	\$2,000	\$2,400	\$400

Under **full information**: both types trade; no market failure.

Under **asymmetric information**: buyers cannot tell lemons from peaches before purchase. A rational buyer pays the *expected* value of a randomly drawn car:

$$E[\text{WTP}] = p \cdot \$2,400 + (1 - p) \cdot \$1,200 = 1,200 + 1,200p. \quad (12.1)$$

where p = fraction of peaches in the market.

12.2.2. The Buyer’s Intuition: Why Peaches Are Driven Out

Before we derive the pooling condition algebraically, pause and think through the logic from the *buyer’s* perspective. This is the heart of Akerlof’s insight.

Imagine you are shopping for a used car. You know that half the cars on the market are lemons (worth \$1,200 to you) and half are peaches (worth \$2,400 to you), but you cannot tell which is which just by looking. What is the most you would rationally pay?

At most, the expected value: $0.5 \times \$1,200 + 0.5 \times \$2,400 = \$1,800$.

But now think about the *seller’s* side. A seller who knows they have a peach values it at \$2,000 (their reservation price). They would *never* accept \$1,800 — they would only sell for at least \$2,000. So at a price of \$1,800, the only cars being offered are lemons. The buyer, anticipating this, realises that any car available at \$1,800 must be a lemon, and so only offers \$1,200. Peach sellers exit the market entirely.

Read this narrative carefully before looking at the algebra. The math captures exactly this logic.

The Adverse Selection Logic in Three Steps.

- (1) Buyers cannot distinguish quality, so they offer only the *average* expected value.
- (2) At that average price, *only low-quality sellers are willing to sell* — high-quality sellers prefer to keep the good rather than accept a below-reservation price.

(3) Buyers, anticipating step (2), revise down their offer. This drives out even more high-quality sellers. The cascade continues until only the lowest quality remains.

This is adverse selection: bad types drive out good types. The presence of low-quality goods imposes a negative externality on high-quality sellers, working entirely through beliefs and prices.

In plain English: “Bad money drives out good” (Gresham’s Law). Akerlof showed the same logic applies to cars, insurance, and much else.

12.2.3. The Pooling/Separating Decision Guide

“Which Equilibrium Am I In?” — The Week 12 Decision Guide

Step 1: Compute $\mathbb{E}[\text{WTP}] = p \cdot (\text{buyer's WTP for peach}) + (1 - p) \cdot (\text{buyer's WTP for lemon})$.

Step 2: Compare $\mathbb{E}[\text{WTP}]$ to the peach seller’s *reservation price* v_s^{peach} :

Condition	Equilibrium	Outcome
$\mathbb{E}[\text{WTP}] \geq v_s^{\text{peach}}$	Pooling	Both types trade; buyers cannot identify quality
$\mathbb{E}[\text{WTP}] < v_s^{\text{peach}}$	Separating	Peach sellers exit; only lemons trade

Critical fraction p^* : set $\mathbb{E}[\text{WTP}] = v_s^{\text{peach}}$ and solve for p :

$$1,200 + 1,200p^* = 2,000 \implies p^* = \frac{800}{1,200} = \frac{2}{3} \approx 0.667.$$

If $p > 2/3$: pooling. If $p < 2/3$: separating (only lemons).

In words: Above p^* , there are enough peaches that buyers will pay a high enough average to keep peach sellers in the market. Below p^* , there aren’t enough peaches and they exit.

Case $p = 0.75$ (mostly peaches). $\mathbb{E}[\text{WTP}] = 1,200 + 900 = \$2,100 > \$2,000$. **Pooling equilibrium:** both types trade at a price between \$2,000 and \$2,100. Lemon sellers receive a cross-subsidy from the pooled price; buyers purchasing lemons overpay.

Case $p = 0.25$ (mostly lemons). $\mathbb{E}[\text{WTP}] = 1,200 + 300 = \$1,500 < \$2,000$. **Separating equilibrium:** peach sellers exit; buyers correctly infer all remaining cars are lemons and pay at most \$1,200. Only lemons trade, at a price between \$1,000 and \$1,200. The \$400 gain-from-trade on every peach transaction is permanently destroyed.

Akerlof’s Core Insight. Informational asymmetry can destroy markets that would function perfectly under full information. The market for peaches fails *not* because peaches are not valuable — the gains from trade are \$400 per car — but because buyers cannot distinguish them from lemons. The presence of low-quality goods imposes a negative externality on high-quality sellers: one that works entirely through *beliefs and prices*, not physical spillovers.

Check Your Understanding

1. If the fraction of peaches rises to $p = 0.90$, does pooling or separating occur in the benchmark model? Verify using the pooling condition.
2. What would happen if buyers could costlessly inspect every car before purchasing? How does this change the adverse selection problem — and what does it imply about the value of transparency?
3. Name one real market (besides used cars) where adverse selection operates. Who is the informed party? Who is the uninformed party?

Using the lemons benchmark (lemon WTP \$1,200/reserve \$1,000; peach WTP \$2,400/reserve \$2,000): (a) verify $p^* = 2/3$; (b) what happens to p^* if buyers value peaches at \$3,000 instead of \$2,400? (c) does a higher buyer valuation for peaches make the market more or less likely to unravel?

12.2.4. Market Unravelling: The Cascade

The logic of adverse selection extends beyond a two-type market. When there are multiple quality levels, the unravelling cascade plays out round by round. In each round, buyers update their beliefs about the remaining pool, the expected value falls, and the next tier of high-quality sellers finds the price unacceptable and exits. The key mechanism is *belief updating*: each exit changes what buyers believe about the remaining pool, which changes the price, which causes the next exit.

Market unravelling: cascade of exits until only the lowest quality remains. Complete failure possible even when every individual transaction is mutually beneficial.

The Three-Round Unravelling Cascade (three-type market: Low 50%, Medium 30%, High 20%)

Round	Pool	$E[WTP]$	Vs. reserve	Who exits
0 (start)	Low+Med+High	$0.5(700) + 0.3(1900) + 0.2(3600) = \$1,640$	$< \$3,000$	High types exit
1	Low+Med (62.5%+37.5%)	$0.625(700) + 0.375(1900) = \$1,150$	$< \$1,500$	Medium types exit
2	Low only: buyers pay $\leq \$700$	—	—	Only lemons remain

Each round of exit reduces average quality, lowers the pooled price, and drives out the next tier. The cascade terminates when only the lowest-quality type remains.

The buyer's belief at each round is what drives the cascade. It is worth tracing these beliefs explicitly:

- **Round 0 (before any exits):** Buyers know all three types exist (Low 50%, Medium 30%, High 20%). Their expected WTP is the weighted average: $0.5(700) + 0.3(1900) + 0.2(3600) = \$1,640$. This is below the High type's reservation price of \$3,000, so High sellers find the pooled price unacceptable and exit.
- **Round 1 (after High exits):** Buyers now know only Low and Medium remain. They update their belief: Low is now $0.5/0.8 = 62.5\%$ of the remaining pool and Medium is $0.3/0.8 = 37.5\%$. The new expected WTP is $0.625(700) + 0.375(1900) = \$1,150$. This

falls below the Medium type's reservation price of \$1,500, so Medium sellers also exit.

- **Round 2 (after Medium exits):** Only Low-quality sellers remain. Buyers' belief collapses: the entire pool is lemons. Buyers offer at most \$700. The market completes its unravelling. Only the lowest-quality type survives.

"Adverse selection only means more bad goods are traded." The deeper harm is that *good goods stop being traded at all*. In the extreme, complete market unravelling destroys every mutually beneficial trade in high-quality goods — even though buyers and sellers would both gain from them. The welfare loss is not from mismatch; it is from the complete elimination of high-quality markets.

Check Your Understanding

1. Lemons: seller \$500, buyer \$700. Peaches: seller \$1,800, buyer \$2,200. Find p^* . Is there a pooling equilibrium at $p = 0.80$? At $p = 0.50$?
2. Why is a pooling equilibrium allocatively *distorted* even when it exists (i.e., when both types trade)?
3. True or false: "Under full information, adverse selection would still cause market failure."

12.2.5. Solutions to Adverse Selection

Signalling (informed party acts)

Signalling: An action taken by the *informed* party (high-quality type) to credibly communicate private information to the uninformed party.

Single-crossing condition (for a signal to work): The marginal cost of the signal must be *lower* for high-quality types than for low-quality types. Then only high-quality types find it worthwhile to signal; low-quality types cannot profitably mimic.

In words: A signal is credible only if it's cheap enough for the high-quality type but too expensive for the low-quality type to copy. The education-as-signal model is the classic application.

Examples: car warranties; education (Spence 1973); brand names; Certified Pre-Owned programmes.

Mathematical Model of Spence Signalling

To formalise signalling, consider a labour market with two types of workers: high-productivity (H) and low-productivity (L). Employers cannot observe productivity directly at hiring but offer wages based on a worker's education level $s \geq 0$ (e.g., years of schooling).

Let worker productivity (and thus their full-information wage) be:

$$\text{Type } H : y_H = \$100,000$$

$$\text{Type } L : y_L = \$40,000$$

The cost of acquiring education level s depends on worker type:

$$c_H(s) = s \quad \text{and} \quad c_L(s) = 3s$$

This cost structure satisfies the **single-crossing condition**: the marginal cost of education is strictly lower for high-productivity workers ($\partial c_H / \partial s = 1 < 3 = \partial c_L / \partial s$). Education does not increase productivity; it is purely a signalling device.

Why the Single-Crossing Condition Is the Key

Before deriving the ICCs algebraically, it is worth understanding geometrically and intuitively why the single-crossing condition is the engine of separating equilibria.

In a diagram with education s on the horizontal axis and wage w on the vertical axis, a worker's indifference curve shows all (s, w) combinations that give the same utility. The slope of the indifference curve at any point is the wage the worker needs to compensate for an additional unit of education — this is exactly the marginal cost of education.

Because the low type's marginal cost ($\partial c_L / \partial s = 3$) is higher than the high type's ($\partial c_H / \partial s = 1$), the low type's indifference curves are *steeper* everywhere in (s, w) space. The two families of indifference curves cross each other exactly once — hence the name “single-crossing.”

This geometric fact has a profound economic implication: there exists an education threshold s^* such that (a) the high type prefers (s^*, w_H) over $(0, w_L)$ — the wage gain exceeds the cost — but (b) the low type does *not* prefer (s^*, w_H) over $(0, w_L)$ — the cost exceeds the wage gain. Education becomes a credible signal because only the high type finds it worth acquiring.

Single-crossing: at any (s, w) , the low type's indifference curve is steeper than the high type's. This means a wage gap Δw can be “bought” by the high type at lower cost.

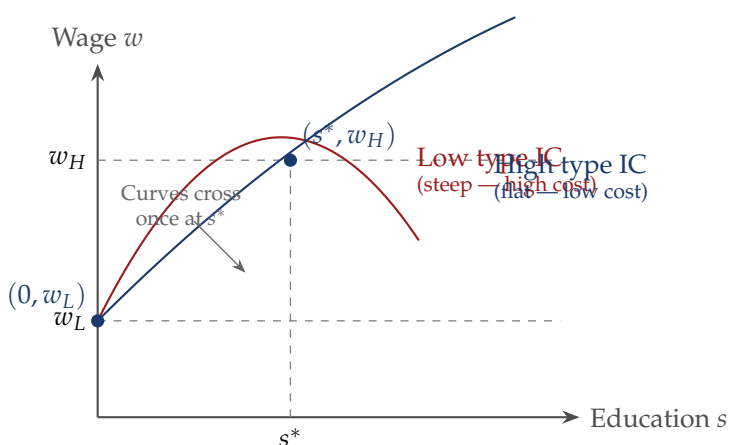


Figure 12.1. Single-crossing of indifference curves. At education level s^* , the high type's indifference curve passes through (s^*, w_H) : the high type is just willing to acquire s^* to earn w_H . The low type's indifference curve is steeper — acquiring s^* would cost the low type more than the wage gain, so the low type stays at $(0, w_L)$. The curves cross exactly once. This single-crossing property is what makes s^* a credible separating signal.

Employers hold beliefs about worker productivity: they pay $w(s) = y_H$ if $s \geq s^*$ and

$w(s) = y_L$ if $s < s^*$, where s^* is a threshold education level.

For a **separating equilibrium** to exist, the threshold s^* must satisfy two Incentive Compatibility Constraints (ICCs):

1. **Low-productivity worker's ICC:** Type L must prefer not to mimic Type H :

$$\begin{aligned} w(0) - c_L(0) &\geq w(s^*) - c_L(s^*) \\ y_L - 0 &\geq y_H - 3s^* \\ 3s^* &\geq y_H - y_L = 100 - 40 = 60 \\ s^* &\geq 20 \end{aligned}$$

2. **High-productivity worker's ICC:** Type H must prefer to signal rather than be treated as Type L :

$$\begin{aligned} w(s^*) - c_H(s^*) &\geq w(0) - c_H(0) \\ y_H - s^* &\geq y_L - 0 \\ s^* &\leq y_H - y_L = 100 - 40 = 60 \end{aligned}$$

Thus, any education threshold s^* in the range:

$$20 \leq s^* \leq 60 \quad (12.3)$$

sustains a separating equilibrium. The **minimum-cost separating equilibrium** (or Riley outcome) occurs at $s^* = 20$, which minimises the deadweight loss of education. Because education is socially unproductive here, any $s^* > 0$ represents pure social waste, demonstrating how asymmetric information can lead to inefficient over-investment in credentialing.

Check Your Understanding

1. Why would it be irrational for a low-type worker to acquire education level $s^* = 30$? Show this using the low type's ICC. (Use the numerical values from the Spence model above.)
2. If the cost difference between types narrows (suppose the low type's cost falls from $3s$ to $2s$), what happens to the range of separating equilibria $[s_{\min}^*, s_{\max}^*]$? Does the market become easier or harder to separate?
3. Is the signalling equilibrium socially efficient? Justify your answer by comparing total output and total resource use under pooling vs. separating.

Why This Matters: Signalling in Business

The Spence signalling model is not just about education. The same single-crossing logic applies throughout business and finance:

- **MBA degrees and professional certifications (CFA, CPA, Bar exam):** Obtaining a CFA charter requires 900+ hours of study. If high-ability finance professionals have lower opportunity costs of study time (because they learn faster or already know more), the certification satisfies single-crossing and serves as a credible signal of ability — even if the curriculum adds little incremental knowledge beyond what the candidate already has.
- **Brand reputation:** Established firms invest in brand equity — advertising, quality consistency, premium pricing. If these investments are more costly for low-quality producers to sustain (they would trigger returns/complaints), they serve as credible quality signals.
- **Product warranties:** A manufacturer that guarantees its product for three years is implicitly signalling confidence in product quality. A low-quality producer offering the same warranty would face high expected warranty claim costs — making mimicry unprofitable.
- **Dividend policy:** Firms with strong cash flows sometimes pay high dividends. Since dividends are costly (they require cash), a financially weak firm cannot credibly mimic — dividends serve as a signal of financial health (the “dividend signalling” model).

The social efficiency question. From the private perspective, signalling restores market efficiency: workers are paid their correct productivity, types are sorted correctly. From the social perspective, the verdict depends on whether the signal has *dual value* — does it also raise productivity? If an MBA programme teaches analytical skills that make graduates more productive (dual value), then signalling is socially efficient. If it purely sorts without adding productivity, then every dollar spent on MBA tuition is socially wasted. Most signals in practice combine both elements, making social efficiency an empirical, not just theoretical, question.

Screening (uninformed party acts)

Signalling requires the informed party to act. Screening is the mirror image: the uninformed party structures the offer to induce self-selection.

The uninformed party designs a **menu of contracts** that induces each type to self-select into the contract designed for it. Classic example: insurance contracts with different deductibles. Low-risk individuals prefer high-deductible/low-premium contracts; high-risk individuals prefer low-deductible/high-premium contracts. The menu separates types through incentive compatibility.

A Screening Menu-of-Contracts Example

Consider an insurance company that cannot observe whether a customer is high-risk (50% chance of a \$10,000 accident) or low-risk (10% chance of the same accident). Equal fractions of each type.

Actuarially fair premiums if the insurer could observe type:

- High-risk: $P_H = 0.50 \times \$10,000 = \$5,000$
- Low-risk: $P_L = 0.10 \times \$10,000 = \$1,000$

But since types are unobservable, the insurer offers a **menu**:

- **Contract A:** Full coverage (\$10,000 payout), premium = \$5,000. Actuarially fair for the high-risk type.
- **Contract B:** Partial coverage (\$4,000 payout), premium = \$400. Actuarially fair for the low-risk type on partial coverage: $0.10 \times \$4,000 = \400 .

Will types self-select correctly? For a risk-neutral customer, the relevant comparison is expected net benefit (expected payout minus premium):

	Contract A	Contract B
High-risk expected benefit	$0.50 \times \$10,000 - \$5,000 = \$0$	$0.50 \times \$4,000 - \$400 = \$1,600$
Low-risk expected benefit	$0.10 \times \$10,000 - \$5,000 = -\$4,000$	$0.10 \times \$4,000 - \$400 = \$0$

Notice the problem: *both* types prefer Contract B as stated! High-risk customers get \$1,600 expected benefit from B vs. \$0 from A. Low-risk customers are indifferent between B (\$0) and not buying.

This illustrates why screening contracts typically involve **distortions**: to make Contract A more attractive to high-risk types relative to low-risk types, the insurer must make Contract B *less* attractive to high-risk types. The standard solution is to reduce coverage in Contract B (the low-risk contract), making it less valuable to high-risk customers who file claims more often.

The Screening Principle. Effective screening menus require distortions: the contract designed for low-risk/low-quality types must be made *less attractive* than the full-information optimum — partial coverage, higher deductibles, or limited quantities — to prevent high-risk/high-quality types from mimicking. The low-risk type bears the cost of information asymmetry through reduced coverage. This is the *second-best* outcome: better than complete market failure, but worse than full-information contracts.

Certification and Reputation

Third-party certification (independent car inspections, credit-rating agencies, professional licensing) converts hidden information into publicly verifiable quality grades. Reputation mechanisms work in repeated interactions: sellers have an incentive to maintain quality because future profits depend on it.

eBay, Ratings, and the Lemon Problem. When eBay launched in 1995, it faced a textbook adverse selection problem: buyers could not inspect goods before purchase. Early economists predicted market unravelling. Instead, eBay pioneered a *reputation/feedback system*: buyers rate sellers after each transaction. Sellers with high ratings charge a premium; those with poor ratings cannot sell at any price. This transforms adverse selection into a repeated game where reputation has direct financial value. Empirical studies consistently find that sellers with 100% positive feedback fetch 5–10% higher prices than otherwise identical listings — precisely the signalling premium theory predicts.

Adverse selection is a pre-contractual problem. We now turn to moral hazard — what happens after the contract is signed.

12.3. Real-World Applications of Adverse Selection

Does adverse selection operate in real markets the way the theory predicts — in health insurance, credit markets, and labour markets? What evidence supports the theory?

12.3.1. Health Insurance: The Death Spiral

Individuals know more about their own health status than any insurer. If the insurer charges a pooled premium equal to average expected cost, healthy individuals — who pay more than their actuarially fair rate — opt out. This raises the average cost of the remaining pool, forcing premiums higher, which drives out the next-healthiest group. In the extreme, only the sickest individuals remain, at very high premiums.

Death spiral: the self-reinforcing exit of low-risk individuals from a pooled insurance market, driving premiums toward the break-even cost of insuring only the highest-risk group.

Application: Health Insurance Adverse Selection and the ACA

The **death spiral** mechanism works as follows:

1. Insurer charges pooled premium $P_{\text{pool}} = \text{average expected claim cost across all potential enrollees}$.
2. Healthy individuals (low expected claims) compare P_{pool} to their expected cost. If $P_{\text{pool}} > \text{expected claim}$, they exit the pool.
3. The remaining pool is now sicker on average. The insurer raises P_{pool} to break even on the new, higher-risk pool.
4. The next-healthiest remaining individuals now face a premium exceeding their expected cost and exit.
5. Repeat: premiums spiral upward, enrollment collapses downward. In the extreme, only the very sick remain at very high premiums — or the market collapses entirely.

The **ACA's individual mandate (2010–2017)** was a direct policy response: compel low-risk individuals to remain in the pool by making non-purchase financially costly. By keeping healthy people enrolled, the mandate stabilises the risk pool and prevents the spiral.

When the mandate's financial penalty was zeroed in 2019, economists tracked markets for evidence of renewed adverse selection dynamics. Many state markets showed rising premiums and insurer exits consistent with the death-spiral prediction — a natural experiment confirming the theory.

Alternative remedies: Risk adjustment (transfers between insurers based on enrollee health status) and reinsurance (subsidising high-cost enrollees) also address adverse selection by reducing the penalty insurers face from attracting sick patients — reducing the incentive to design plans that deter high-risk enrollees.

12.3.2. Credit Markets: Stiglitz–Weiss Credit Rationing

Banks face adverse selection: borrowers know the riskiness of their projects better than lenders. To understand how this leads to credit rationing, consider a bank lending to entrepreneurs who require $L = \$1,000$ in capital. There are two types of projects in the market, present in equal proportions:

- **Safe Project (S):** Succeeds with probability $p_S = 0.9$. If successful, it yields a return of

$R_S = \$1,500$; if it fails, it yields \$0. Expected return: $0.9 \times \$1,500 = \$1,350$.

- **Risky Project (R):** Succeeds with probability $p_R = 0.5$. If successful, it yields a return of $R_R = \$2,500$; if it fails, it yields \$0. Expected return: $0.5 \times \$2,500 = \$1,250$.

Entrepreneurs have limited liability (they repay only upon success) and must repay $L(1 + r)$ to the bank, where r is the interest rate.

The expected profit for entrepreneur type $i \in \{S, R\}$ is:

$$\mathbb{E}[\Pi_i(r)] = p_i[R_i - L(1 + r)]$$

Evaluating this for both types:

$$\mathbb{E}[\Pi_S(r)] = 0.9[1,500 - 1,000(1 + r)] = 450 - 900r$$

$$\mathbb{E}[\Pi_R(r)] = 0.5[2,500 - 1,000(1 + r)] = 750 - 500r$$

Notice that safe types are more sensitive to the interest rate because they succeed more often ($\partial \mathbb{E}[\Pi_S]/\partial r = -900 < -500 = \partial \mathbb{E}[\Pi_R]/\partial r$).

Safe entrepreneurs will only apply for loans if their expected profit is non-negative:

$$450 - 900r \geq 0 \implies r \leq 0.50 \text{ (or 50\%)}$$

Let $r^* = 0.50$ be the critical interest rate.

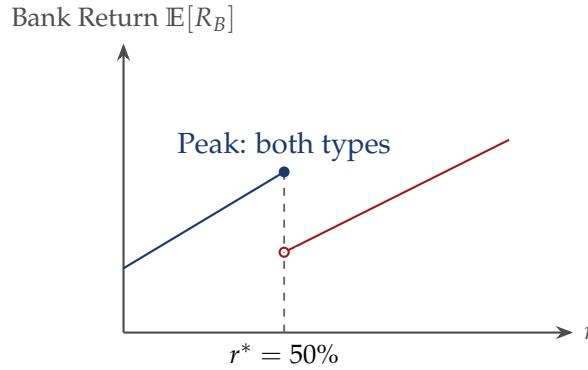


Figure 12.2. Non-monotonic bank returns and credit rationing. When the interest rate r rises above $r^* = 50\%$, safe borrowers exit the market. The bank's expected return drops discontinuously because the borrower pool becomes entirely comprised of risky types (adverse selection).

The expected return to the bank per loan offered depends on who applies:

1. **Pooling Pool ($r \leq 50\%$):** Both types apply. The bank's expected return is:

$$\mathbb{E}[R_B(r)] = \frac{1}{2}p_S L(1 + r) + \frac{1}{2}p_R L(1 + r) = 0.7 \times 1,000(1 + r) = 700(1 + r)$$

At $r = 50\%$, the bank's expected return is $700(1.5) = \$1,050$.

2. **Risky-Only Pool ($r > 50\%$):** Safe types exit. The bank's expected return is:

$$\mathbb{E}[R_B(r)] = p_R L(1 + r) = 0.5 \times 1,000(1 + r) = 500(1 + r)$$

Just above $r = 50\%$, the bank's return falls to \$750, and only rises back to \$1,050 if the bank increases the rate to $r = 110\%$.

Because raising the interest rate above 50% drives out the low-risk borrowers, the bank's expected return is non-monotonic and exhibits a sharp drop at $r^* = 50\%$. If the bank faces excess demand for loans at $r = 50\%$, it will refuse to raise the interest rate to clear the market. Doing so would worsen the quality of the applicant pool and lower its expected profits. Instead, the bank rations credit, allocating loans by quantity rather than price.

12.3.3. Labour Markets: Spence Signalling

Employers cannot observe worker productivity at the time of hiring. They offer wages based on average productivity, undervaluing high-productivity workers who then invest in education to signal their type. Low-productivity workers, overvalued by the average wage, flood the market. The result: over-representation of low-productivity workers and socially wasteful investment in signalling by high-productivity types.

Check Your Understanding

1. Describe the death spiral in health insurance. What is the equilibrium if the mandate is removed?
2. Why does raising the interest rate sometimes *reduce* bank profit? Which assumption of standard supply/demand breaks down?
3. In Spence's signalling model, education doesn't make workers more productive. Why do high-ability workers still get it?

12.4. Moral Hazard: Hidden Action

A CEO earns a flat \$3 million salary regardless of firm performance. What effort does she exert? What contract would make her choose the socially optimal effort level?

Moral Hazard (Hidden Action Problem): After a contract is signed, the **agent** takes an action that affects the outcome of the contract but that the **principal** cannot perfectly observe or verify. Because the agent does not bear the full consequences of her actions, she under-supplies effort.

Timing: *ex post* (post-contractual). The asymmetry emerges after the deal is struck.

Examples: After purchasing comprehensive car insurance, a driver takes less care (insurer cannot monitor daily behaviour). Corporate executives pursue perquisites rather than shareholder value (shareholders cannot observe every CEO decision). Employees on fixed hourly wages shirk when direct supervision is costly.

Moral hazard is hidden action after contracting. Adverse selection is hidden type before contracting.

12.4.1. The Principal-Agent Model

Canonical setup:

- (1) **Stage 1 (Contract):** The **principal** (shareholder, employer) designs a compensation contract $w(\pi)$, where π is verifiable profit.
- (2) **Stage 2 (Effort):** The **agent** (CEO, worker) accepts or rejects; if accepted, chooses hidden effort e to maximise personal utility $U = w - c(e)$.

Profit $\pi = \pi(e, \varepsilon)$ depends on effort and a random shock. The principal writes contracts on π , not on e .

12.4.2. Three Compensation Schemes

Benchmark: $\pi = 20e$, $c(e) = 2e^2$, outside option $\bar{U} = 20$.

Check Your Understanding

1. **Prediction exercise (before the math):** The firm offers the CEO one of three contracts: (S_1) a fixed salary of \$50 regardless of profit; (S_2) a base salary of \$20 plus 50% of profit; (S_3) zero base salary plus 100% of profit (i.e., the CEO effectively “buys” the firm from shareholders for a fixed fee). Before working through the math: which contract do you predict maximises the CEO’s effort? Which maximises the firm’s profit net of compensation? Which is best for a *risk-averse* CEO?
2. What is the CEO’s effort under a flat salary? What does your answer imply about the connection between pay and accountability?
3. If the principal could directly observe effort e , would she need to offer performance pay at all? What would the optimal contract look like?

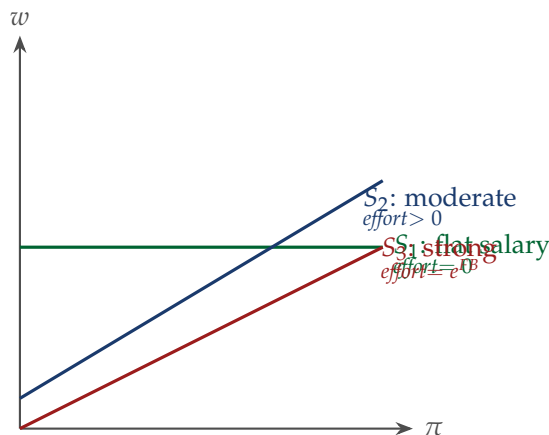


Figure 12.3. Three compensation schemes. What to look for: Steeper slope = stronger incentive = higher effort. Under S_1 (flat), pay is independent of profit so effort = 0. Under S_3 (full residual claim, slope = 1), the CEO internalises all profit and exerts first-best effort.

How the Pay Structure Drives Effort

Scheme	Contract	CEO's FOC	Effort
S_1 : Fixed salary	$w = \$3\text{M}$ (constant)	None	$e = 0$
S_2 : Moderate link	$w = w_0 + 0.5\pi$	$10\beta - 4e = 0$	$e = 2.5$
S_3 : Full residual claim	$w = w_0 + \pi$ ($\beta = 1$)	$20 - 4e = 0$	$e = 5 = e^{FB}$

General rule: CEO's

FOC is $20\beta - 4e = 0 \Rightarrow e^*(\beta) = 5\beta$. At $\beta = 1$: first-best $e^{FB} = 5$ is achieved.

Answer to prediction exercise: (S_3) maximises CEO effort (the CEO internalises all profit) and firm profit net of compensation (shareholders receive the fixed fee). A risk-averse CEO prefers (S_1) (no income risk) — but this produces zero effort. This is the incentive–insurance trade-off in preview.

12.4.3. Solving for the Optimal Contract

With $\beta = 1$ and binding participation constraint:

$$w_0 + 50\beta^2 = \bar{U} \Rightarrow w_0 = 20 - 50(1)^2 = -30.$$

At $\beta^* = 1$: $e^* = 5$; $\pi = 100$; CEO pay = $(-30) + 100 = \$70$; shareholders keep $100 - 70 = \$30$. Total surplus = $\$50 = 20(5) - 2(25)$. ✓

In words: The principal “sells the firm” to the agent for a lump sum of \$30 (the negative fixed fee). The agent keeps all profit and chooses first-best effort. This perfectly aligns incentives.

12.4.4. The Incentive–Insurance Trade-off and Optimal β^*

High-powered contracts solve the effort problem but create a risk problem: a risk-averse CEO exposed to income variance demands a *risk premium*. The optimal contract balances:

Optimal Performance Weight (Holmström–Milgrom):

$$\beta^* = \frac{1}{1 + r\sigma^2} \quad (12.2)$$

where r = agent's coefficient of absolute risk aversion and σ^2 = variance of profit shock ε .

Mathematical Derivation of the Optimal β^*

To understand the trade-off, we derive the optimal incentive contract. Assume a risk-neutral principal (shareholders) and a risk-averse agent (CEO) with CARA utility and cost of effort $c(e) = \frac{1}{2}e^2$.

The firm's profit is $\pi = e + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$. The principal offers a linear contract $w = w_0 + \beta\pi$.

The agent's utility can be represented by the **Certainty Equivalent (CE)** of the contract net of effort cost:

$$CE = \mathbb{E}[w] - c(e) - \frac{1}{2}r\text{Var}(w)$$

Given $w = w_0 + \beta(e + \varepsilon)$, we have $\mathbb{E}[w] = w_0 + \beta e$ and $\text{Var}(w) = \beta^2\sigma^2$. Substituting these yields:

$$CE = w_0 + \beta e - \frac{1}{2}e^2 - \frac{1}{2}r\beta^2\sigma^2$$

The agent chooses effort e to maximise CE . The first-order condition is:

$$\frac{\partial CE}{\partial e} = \beta - e = 0 \implies e^*(\beta) = \beta$$

To induce the agent to accept the contract, the principal must satisfy the Participation Constraint (PC), which requires $CE \geq \bar{U}$. At the profit-maximising contract, the PC must bind ($CE = \bar{U}$):

$$w_0 = \bar{U} - \beta e^* + \frac{1}{2}(e^*)^2 + \frac{1}{2}r\beta^2\sigma^2 = \bar{U} - \frac{1}{2}\beta^2 + \frac{1}{2}r\beta^2\sigma^2$$

The risk-neutral principal maximises expected profit net of compensation:

$$\begin{aligned} \max_{\beta} \quad & \mathbb{E}[\pi] - \mathbb{E}[w] \\ \max_{\beta} \quad & e^*(\beta) - (w_0 + \beta e^*(\beta)) \end{aligned}$$

Substituting the binding w_0 and $e^*(\beta) = \beta$:

$$\max_{\beta} \quad \beta - \frac{1}{2}\beta^2 - \frac{1}{2}r\beta^2\sigma^2 - \bar{U}$$

Taking the first-order condition with respect to β :

$$1 - \beta - r\beta\sigma^2 = 0 \implies \beta^* = \frac{1}{1 + r\sigma^2}$$

This elegant formula shows that the optimal contract is shaped by the incentive-insurance trade-off: when noise (σ^2) or risk aversion (r) is zero, $\beta^* = 1$ (the agent bears all risk and exerts first-best effort). As risk aversion or noise increases, the contract shifts toward insurance (lower β), weakening effort incentives to reduce the risk premium.

Comparative Statics: How Risk and Noise Shape the Optimal Contract

The formula $\beta^* = 1/(1 + r\sigma^2)$ has two important comparative statics. It is worth understanding the intuition behind each, not just the direction.

Comparative statics: how the optimal answer changes when one parameter changes, all else equal.

As r (the agent's risk aversion) rises, β^* falls. A more risk-averse agent dislikes income variability. Under a high- β contract, the agent's pay swings with profit — which includes the random shock ε . A very risk-averse CEO would demand a large risk premium to accept this income volatility, making the contract expensive for the principal. The optimal response is to reduce β : accept somewhat lower effort (a cost) in exchange for a lower risk premium (a saving). The balance between these two — incentives vs. insurance — determines β^* .

As σ^2 (output variance/noise) rises, β^* falls. In a noisy environment, profit is a poor signal of effort. Even if the CEO works hard, profit may be low (bad luck) or high (good luck) regardless. Tying pay tightly to a noisy profit signal imposes income risk on the CEO without proportionate benefit in incentive provision. The principal responds by reducing β : rely less on the noisy signal, provide more insurance, and accept somewhat weaker incentives.

Numerical example. Suppose $r = 0.02$ and $\sigma^2 = 100$:

$$\beta^* = \frac{1}{1 + 0.02 \times 100} = \frac{1}{1 + 2} = \frac{1}{3} \approx 0.33.$$

If risk aversion doubles to $r = 0.04$:

$$\beta^* = \frac{1}{1 + 0.04 \times 100} = \frac{1}{1 + 4} = \frac{1}{5} = 0.20.$$

A doubling of risk aversion reduces the optimal performance weight from $1/3$ to $1/5$ — a 40% reduction. The CEO's pay becomes much more heavily weighted toward the fixed salary w_0 and much less toward profit sharing.

Condition	β^*	Interpretation
$r = 0$ (risk-neutral agent)	$\beta^* = 1$	First-best achievable
$\sigma^2 = 0$ (no noise)	$\beta^* = 1$	First-best achievable
$r \uparrow$ (more risk-averse)	$\beta^* \downarrow$	Shift toward insurance
$\sigma^2 \uparrow$ (more noise)	$\beta^* \downarrow$	Shift toward insurance

Application: CEO Compensation in Practice

Real-world CEO pay packages at S&P 500 companies typically combine three elements, each corresponding to a piece of the principal–agent model:

- **Base salary (fixed, w_0):** Provides a guaranteed floor — the “insurance” component. Ensures the CEO meets her participation constraint. In the model, a higher w_0 allows the principal to set a lower β while still satisfying the PC.
- **Annual bonus (tied to accounting profits or earnings per share):** This is the short-term performance-pay component — the $\beta \cdot \pi$ term in the model. It provides incentives for near-term profit maximisation but creates multi-tasking risk: CEOs may manage earnings (sacrifice long-term value for short-term EPS).
- **Long-term equity grants (stock options, restricted stock):** The longer-horizon performance component. Options are valuable only if the stock price rises above a threshold — creating high-powered incentives aligned with long-run shareholder value. The vesting period (typically 3–5 years) addresses the horizon problem.

Industry variation in β . The model predicts that β^* should be *lower* in industries with higher uncertainty (σ^2). This matches the evidence:

Industry	Output uncertainty σ^2	Predicted β^*
Regulated utilities	Low (stable revenue)	High
Financial services	Medium	Medium
Biotech / startups	Very high (R&D outcomes)	Low

Empirically, utility CEOs receive a larger fraction of pay as salary (low β , high insurance), while startup CEOs receive mostly equity (high β — but also high noise, which theory says should lower β^*). The equity-heavy startup contract is rationalised by the fact that startup CEOs are also major shareholders, making monitoring easier and risk tolerance higher than the basic model assumes.

12.4.5. Limits of High-Powered Contracts

Multi-tasking (Holmström–Milgrom 1991): When agents perform multiple tasks, tying pay to one measurable metric causes over-investment in the measured activity and neglect of unmeasured but equally important ones. A teacher evaluated only on test scores neglects students’ well-being. A trader compensated on short-term returns takes excessive tail risk.

Horizon problem: Managers nearing retirement underinvest in long-term projects. Equity-based compensation with vesting periods is one response.

Earnings manipulation: If pay is tied to reported profits, managers have incentives to manipulate accounting figures. Independent auditing and boards are partial remedies.

Wells Fargo Cross-Selling Scandal (2016). Employees were evaluated on the number of new accounts opened per customer — a high-powered incentive. The result: millions of unauthorised accounts opened without customers’ knowledge. The bank paid \$3 billion in settlements and fired 5,000+ employees. This is precisely the multi-tasking distortion: when one dimen-

sion of a complex job is measured and rewarded, agents neglect unmeasured dimensions and, in extremes, game the metric entirely. Sometimes the solution is a *flatter* contract, not a steeper one.

Check Your Understanding

1. CEO benchmark: $\pi = 20e$, $c(e) = 2e^2$, $\bar{U} = 20$. At $\beta = 0.5$: what is e^* ? At $\beta = 1$: what is e^* ? What is first-best e^{FB} ?
2. From the formula $\beta^* = 1/(1 + r\sigma^2)$: what happens to the optimal contract if profit becomes more volatile? Why? Illustrate using $r = 0.02$ and σ^2 doubling from 100 to 200.
3. True or false: "If the agent is risk-neutral, the principal can always achieve first-best." Explain any additional conditions required.

12.4.6. Moral Hazard with Discrete Effort and Risk Aversion

In many economic models, effort is discrete rather than continuous. Consider a principal-agent game where the agent chooses between two effort levels: High (e_H) and Low (e_L).

The agent is risk-averse with utility function $U(w, e) = \sqrt{w} - c(e)$, where the cost of effort is $c(e_H) = 4$ and $c(e_L) = 0$. The agent's reservation utility is $\bar{U} = 4$.

Firm profit can be high ($\pi_H = \$200$) or low ($\pi_L = \50). The probability distribution of profit depends on effort:

- Under high effort (e_H): $\Pr(\pi_H) = 0.8$, $\Pr(\pi_L) = 0.2$
- Under low effort (e_L): $\Pr(\pi_H) = 0.3$, $\Pr(\pi_L) = 0.7$

If effort is observable (the **First-Best** benchmark), the principal can contract on effort directly. To implement e_H , the principal offers a fixed wage w on the condition that the agent exerts high effort. The participation constraint binds:

$$\sqrt{w^{FB}} - 4 = 4 \implies \sqrt{w^{FB}} = 8 \implies w^{FB} = \$64$$

If effort is unobservable (the **Second-Best** case), the principal must write a contract (w_H, w_L) contingent on profits to satisfy two constraints:

1. Participation Constraint (PC):

$$0.8\sqrt{w_H} + 0.2\sqrt{w_L} - 4 \geq 4 \implies 0.8\sqrt{w_H} + 0.2\sqrt{w_L} \geq 8$$

2. Incentive Compatibility Constraint (ICC):

$$0.8\sqrt{w_H} + 0.2\sqrt{w_L} - 4 \geq 0.3\sqrt{w_H} + 0.7\sqrt{w_L} - 0 \implies 0.5\sqrt{w_H} - 0.5\sqrt{w_L} \geq 4$$

At the cost-minimising optimum, both constraints bind. Let $u_H = \sqrt{w_H}$ and $u_L = \sqrt{w_L}$.

PC: agent is willing to participate. ICC: agent finds it optimal to exert the level of effort the principal desires. The PC and ICC serve different purposes:

We solve the system:

$$\begin{aligned} 0.8u_H + 0.2u_L &= 8 \\ 0.5u_H - 0.5u_L &= 4 \implies u_H - u_L = 8 \implies u_H = u_L + 8 \end{aligned}$$

Substituting u_H into the PC:

$$0.8(u_L + 8) + 0.2u_L = 8 \implies u_L + 6.4 = 8 \implies u_L = 1.6$$

Then, $u_H = 1.6 + 8 = 9.6$. Finding the wages:

$$\begin{aligned} w_L^* &= (1.6)^2 = \$2.56 \\ w_H^* &= (9.6)^2 = \$92.16 \end{aligned}$$

The expected wage cost to implement high effort under asymmetric information is:

$$\mathbb{E}[w^*] = 0.8 \times \$92.16 + 0.2 \times \$2.56 = \$73.728 + \$0.512 = \$74.24$$

Since the first-best wage is $w^{FB} = \$64$, the **cost of asymmetric information** (the incentive rent and risk premium paid to the agent) is $\$74.24 - \$64.00 = \$10.24$.

12.5. Comparing Adverse Selection and Moral Hazard

We have now studied two information failures in depth. How do we quickly tell them apart in a new context — and what does each analysis imply for policy?

Adverse Selection vs. Moral Hazard: The Complete Comparison

Feature	Adverse Selection	Moral Hazard
Hidden variable	Type / quality (fixed characteristic)	Effort / action (chosen after contract)
Timing	Pre-contractual (<i>ex ante</i>)	Post-contractual (<i>ex post</i>)
Who holds info	Agent knows her own type	Agent chooses unobservable action
Market failure	High-quality goods crowded out; market unravelling	Sub-optimal effort; inefficient risk allocation
Remedies	Signalling, screening, certification, mandates	Performance pay, monitoring, deductibles, co-pays
Classic example	Akerlof's lemons model (1970)	Principal-agent / CEO compensation
Nobel Prize	Akerlof, Spence, Stiglitz (2001)	Holmström, Hart (2016)

The Mnemonic Test
Lie-detector test (reveals private info at contracting): eliminates **adverse selection** — types are now known.
Camera (monitors all actions after contracting): eliminates **moral hazard** — effort is now observable.

Confusing adverse selection with moral hazard. The distinction is timing and what is hidden. **Adverse selection:** hidden types that exist *before* contracting — you cannot change your pre-existing health risk. **Moral hazard:** hidden actions *chosen after* contracting — you choose how carefully you drive after buying insurance.

Use the mnemonic: a lie-detector test would fix adverse selection; a camera would fix moral hazard.

Check Your Understanding

1. Apply the lie-detector/camera test to each scenario and identify the type of information problem. Briefly justify your answer.
 - (a) A car mechanic quotes unnecessary repairs because the customer cannot verify what the car actually needs.
 - (b) A job applicant exaggerates credentials on a CV that employers cannot easily verify at the hiring stage.
 - (c) A health insurance customer becomes less careful about diet and exercise after obtaining comprehensive coverage.
 - (d) A used laptop seller knows about a cracked internal frame that is invisible to the buyer.
2. For each scenario in (1), name one market remedy — drawn from the tools studied in this chapter — that would reduce the problem.
3. True or false: “A problem that can be solved by a camera but not a lie-detector is always a moral hazard.” Explain carefully, and give a scenario where both tools

would help.

Where Does This Take Us?

- **Contract theory and mechanism design:** The principal–agent model is the foundation of mechanism design — designing rules and incentives to elicit desired behaviour from privately informed agents. This is central to auction design, regulation, and institutional economics.
- **Incomplete contracts (Hart–Moore):** When contingencies cannot be written into a contract, parties may underinvest in relationship-specific assets. This is a deeper form of the moral hazard problem that explains firm boundaries and ownership structures.
- **Market design:** Matching markets (job markets, school choice, kidney exchange) are partly designed to solve adverse selection by creating credible signalling mechanisms or centralised screening procedures.
- **Regulatory economics:** Utility regulation faces both adverse selection (the regulator doesn’t know the firm’s true costs) and moral hazard (the firm chooses unobservable cost-reduction effort). Regulatory contract design mirrors the principal–agent framework.
- **Behavioural economics:** Agents may not behave as the rational model predicts — they may underestimate moral hazard temptations (optimism bias) or fail to signal effectively. These deviations are studied in behavioural contract theory.

Seven Common Exam Mistakes — and How to Avoid Them

1. **Confusing adverse selection with moral hazard.** The diagnostic is *timing*. Adverse selection: the information gap exists *before* the contract is signed (ex ante hidden type). Moral hazard: the information gap emerges *after* the contract is signed (ex post hidden action). Use the mnemonic: lie-detector → adverse selection; camera → moral hazard.
2. **Using the buyer’s WTP to compute p^* instead of the seller’s reservation price.** The pooling condition is $\mathbb{E}[\text{WTP}] \geq v_s^{\text{peach}}$ (the seller’s minimum acceptable price), not the buyer’s maximum. For the benchmark: $p^* = (2,000 - 1,200) / (2,400 - 1,200) = 2/3$. The seller’s reservation price \$2,000 is the threshold; the buyer’s WTP \$2,400 appears only inside the expectation.
3. **Evaluating the pooling condition at the wrong fraction.** p^* is the *minimum* fraction of peaches needed for a pooling equilibrium to exist — not the fraction at which pooling “just barely fails.” Pooling occurs when $p \geq p^*$. If the question gives you $p = 0.80$ and $p^* = 2/3$, pooling occurs; if $p = 0.50 < p^*$, only lemons trade. Do not reverse the inequality.
4. **Setting $\beta = 1$ as optimal without checking risk aversion.** When the agent is risk-averse, $\beta^* = 1 / (1 + r\sigma^2) < 1$. First-best ($\beta = 1$) is only optimal if the agent is risk-neutral ($r = 0$) or if profit is fully deterministic ($\sigma^2 = 0$). Always check whether the agent bears income risk before asserting $\beta^* = 1$.
5. **Confusing the Participation Constraint (PC) and the Incentive Compatibility Constraint (ICC).** These serve entirely different purposes. The PC ensures the agent is willing to accept the contract at all: expected utility $\geq \bar{U}$ (the outside option). The ICC ensures the agent, having accepted, finds it optimal to take the *action* the principal wants (high effort, or the

“right” type). A contract can satisfy the PC (the agent accepts) while violating the ICC (the agent then shirks). Both must hold simultaneously.

6. **Forgetting that the participation constraint must bind at the optimum.** The principal minimises expected wage cost subject to PC and ICC. At the optimum, the PC binds (the agent earns exactly $\bar{U}$). If you leave the PC non-binding, you are paying the agent more than necessary — which the principal would not do.
7. **Claiming signalling always improves social welfare.** Signalling restores private market efficiency (correct wages, correct sorting) but often destroys social value. Education that is purely a signal (teaches nothing productive) is socially wasteful: total output is unchanged, but resources are spent on the signal. The signalling equilibrium can be Pareto dominated by a pooling equilibrium where nobody signals. Ask whether education (or any signal) actually raises productivity before concluding that signalling is socially beneficial.

12.6. Solved Examples

Example 12.1: Pooling vs. Separating Equilibrium.

Lemons: seller \$800, buyer \$1,000. Peaches: seller \$3,000, buyer \$3,600.

(a) $p = 0.60$: $E[\text{WTP}] = 0.6(3,600) + 0.4(1,000) = \$2,560 < \$3,000$. Peach sellers exit. **Separating equilibrium.** Only lemons.

(b) $p = 0.30$: $E[\text{WTP}] = 0.3(3,600) + 0.7(1,000) = \$1,780 < \$3,000$. **Separating.**

(c) Knife-edge p^* : $3,600p^* + 1,000(1 - p^*) = 3,000 \Rightarrow 2,600p^* = 2,000 \Rightarrow p^* \approx 0.769$. At least 76.9% peaches required for pooling.

Example 12.2: Optimal CEO Contract.

$\pi = 10e$, $c(e) = e^2$, $\bar{U} = 10$, CEO risk-neutral.

(a) CEO FOC: $10\beta - 2e = 0 \Rightarrow e^*(\beta) = 5\beta$.

(b) PC binds: $w_0 + 25\beta^2 = 10 \Rightarrow w_0 = 10 - 25\beta^2$. Shareholders' profit: $\Pi_S = 50\beta - 10 - 25\beta^2$. FOC: $50 - 50\beta = 0 \Rightarrow \beta^* = 1$.

(c) At $\beta^* = 1$: $e^* = 5$, $\pi = 50$, $w_0 = -15$. CEO pay = $-15 + 50 = \$35$. Shareholders: $50 - 35 = \$15$. Total surplus = $50 - 25 = \$25$. ✓

Example 12.3: Insurance Adverse Selection.

Careful drivers: $q_C = 0.1$; Reckless: $q_H = 0.4$; loss \$10,000; equal fractions.

(a) Pooled premium: $\bar{q} = 0.25$; $P = \$2,500$.

(b) Careful driver's expected loss: $0.1 \times \$10,000 = \$1,000 < \$2,500$. Careful drivers exit. **Adverse selection death spiral.**

(c) Only reckless remain. Break-even premium: \$4,000.

(d) Screening: offer two contracts — full coverage at \$4,000 (reckless self-select) and partial coverage with high deductible at low premium (careful self-select). The deductible is costlier for reckless drivers (triggered more often), creating separation.

Chapter at a Glance — Week 12: Information as Market Failure

Your one-page revision guide.

Adverse Selection — Key Formulas

$$\mathbb{E}[\text{WTP}] = p \cdot \text{WTP}_H + (1 - p) \cdot \text{WTP}_L$$

$$p^* \text{ s.t. } \mathbb{E}[\text{WTP}] = v_s^{\text{peach}}$$

$$\mathbb{E}[\text{WTP}] \geq v_s^{\text{peach}}$$

$$\mathbb{E}[\text{WTP}] < v_s^{\text{peach}}$$

Expected buyer value

Knife-edge fraction

Pooling equilibrium

Separating (only lemons)

Lemons Benchmark ($\text{WTP}_H = 2,400$, $\text{WTP}_L = 1,200$, $v_s = 2,000$)

$$\mathbb{E}[\text{WTP}] = 1,200 + 1,200p; \quad p^* = 2/3$$

$$p = 0.75 \rightarrow \$2,100 > 2,000: \text{pooling } \checkmark$$

$$p = 0.25 \rightarrow \$1,500 < 2,000: \text{separating } \checkmark$$

Spence Signalling (ICCs)

$$c_H(s) = s, \quad c_L(s) = 3s; \quad \text{separating range: } 20 \leq s^* \leq 60$$

$$\text{ICCL: } s^* \geq (y_H - y_L)/3 = 20; \quad \text{ICCH: } s^* \leq y_H - y_L = 60$$

CEO Contract Benchmark ($\pi = 20e$, $c = 2e^2$, $\bar{U} = 20$)

$$e^*(\beta) = 5\beta; \quad e^{FB} = 5 \text{ at } \beta^* = 1$$

$$\text{PC: } w_0 = 20 - 50\beta^2$$

$$\text{Shareholders: } 100\beta - 20 - 50\beta^2; \quad \beta^* = 1$$

$$\text{At } \beta^* = 1: w_0 = -30, \text{ CEO pay} = 70, \text{ shareholders} = 30 \checkmark$$

Optimal β^* and Comparative Statics

$$\beta^* = \frac{1}{1 + r\sigma^2}: \quad r \uparrow \Rightarrow \beta^* \downarrow; \quad \sigma^2 \uparrow \Rightarrow \beta^* \downarrow; \quad r = 0 \Rightarrow \beta^* = 1 \text{ (first-best)}$$

$$\text{Example: } r = 0.02, \sigma^2 = 100 \Rightarrow \beta^* = 1/3; \quad r = 0.04 \Rightarrow \beta^* = 1/5$$

AS vs. MH — Quick Reference

	Hidden?	Type	Adv. Sel.	Moral Hazard
Timing	Ex ante	Ex ante	Ex ante	Ex post
Test	Lie detector	Lie detector	Lie detector	Camera
Fix	Signal/screen	Signal/screen	Signal/screen	Perf. pay
Nobel	2001	2001	2001	2016

PC vs. ICC — Key Distinction

- **PC:** agent accepts the contract ($CE \geq \bar{U}$)
- **ICC:** agent takes the desired action / reveals type
- Both must hold; both bind at the optimum

LO Checklist

- LO1: $\mathbb{E}[\text{WTP}]$ formula; find $p^* \checkmark$
- LO2: Trace unravelling cascade with beliefs $\checkmark$
- LO3: Signalling (single-crossing, Spence ICCs); screening $\checkmark$
- LO4: CEO contract: $e^*(\beta)$, first-best $\checkmark$
- LO5: PC and ICC; solve for β^* ; comparative statics $\checkmark$
- LO6: Timing test; lie-detector/camera $\checkmark$

Solutions to AS

- Signalling: informed party; single-crossing needed
- Screening: uninformed party; menu of contracts (with distortions)
- Certification: third party; converts to public info
- Mandates: compel participation (health insurance)

The thread: Asymmetric information $\rightarrow$ adverse selection (ex ante hidden type): buyers pay average value $\rightarrow$ peach sellers exit if $p < p^* \rightarrow$ unravelling cascade possible (belief updating each round) $\rightarrow$ remedies: signalling (single-crossing, Spence ICCs), screening (menu with distortions), certification $\rightarrow$ moral hazard (ex post hidden action): fixed salary $\rightarrow$ zero effort; profit-sharing $\rightarrow$ effort rises; $\beta^* = 1$ achieves first-best if risk-neutral and no noise $\rightarrow$ risk aversion $\rightarrow$ incentive-insurance trade-off $\rightarrow$ $\beta^* = 1/(1 + r\sigma^2)$: $r \uparrow$ or $\sigma^2 \uparrow \rightarrow \beta^* \downarrow \rightarrow$ multi-tasking: high-powered contracts distort effort across tasks. **Test:** lie-detector fixes AS; camera fixes MH. PC ensures acceptance; ICC ensures desired action.

12.7. Key Terms

Adverse Selection

Pre-contractual (ex ante) failure: hidden type / quality drives high-quality goods out. Also called the hidden information problem.

Lemons Model

Akerlof (1970): informational asymmetry about car quality; lemons crowd out peaches.

Pooling Equilibrium

Both types trade at a single pooled price; buyers cannot identify quality.

Separating Equilibrium

Only low-quality goods trade; high-quality sellers cannot obtain their reservation price.

Market Unravelling

Cascade of exits as each round of high-quality exit lowers average quality and the pooled price. Driven by buyer belief updating at each round.

Critical Fraction p^*

Fraction of peaches at which $\mathbb{E}[\text{WTP}] = v_s^{\text{peach}}$; pooling iff $p \geq p^*$.

Signalling

Informed party takes a costly action to credibly reveal type. Requires single-crossing: signal cheaper for high-quality types.

Single-Crossing Condition

Marginal cost of signal is lower for high-quality types, making mimicry unprofitable for low-quality types. Implies indifference curves cross exactly once.

Screening

Uninformed party designs a menu of contracts inducing each type to self-select via incentive compatibility. Requires distortions to prevent mimicry.

Moral Hazard

Post-contractual (ex post) failure: agent takes hidden action affecting outcome; effort

is sub-optimal. Also called the hidden action problem.

Principal-Agent Model

Principal designs contract; agent takes hidden action. Contract written on verifiable π , not on unverifiable e .

Participation Constraint (PC)

Agent's expected utility $\geq \bar{U}$. Ensures the agent accepts the contract. Binds at the optimum.

Incentive Compatibility (ICC)

Agent finds it optimal to take the action the principal desires (high effort, or reveal true type). Distinct from the PC.

High-Powered Incentive Contract

Pay strongly tied to performance metric; agent becomes (partial) residual claimant.

Incentive-Insurance Trade-off

Higher β induces more effort but exposes risk-averse agent to income risk and demands a higher risk premium.

Optimal β^*

$\beta^* = 1/(1 + r\sigma^2)$. First-best ($\beta^* = 1$) iff agent is risk-neutral or profit is deterministic. Falls with r or σ^2 .

Multi-tasking Problem

Contracting on one task crowds out effort on unmeasured tasks. Calls for flatter, not steeper, contracts.

Credit Rationing

Banks ration credit at a below-clearing rate rather than raise rates and trigger adverse selection (Stiglitz-Weiss 1981).

Residual Claimant

Agent who keeps all (or a fixed share of) output; eliminates effort distortion but shifts all income risk onto the agent.

Death Spiral

Self-reinforcing exit of low-risk individuals from a pooled insurance market, driven by rising premiums that price out the healthiest remaining enrollees.

Recommended Videos

Four videos selected for clarity and direct relevance.

1. Adverse Selection and the Lemons Problem (*Academy 4 Social Civics*) ~ 10 min

<https://www.youtube.com/watch?v=IOMCLYBQpgg>

Derives the lemons model, explains the pooling/separating distinction, and traces the unravelling argument. Recommended before Level 1–2 adverse selection questions.

2. Adverse Selection, Signalling, and Screening (*Solomon Geleta*) ~ 12 min

<https://www.youtube.com/watch?v=nxabqQK-4pg>

Covers signalling (single-crossing), screening (menu of contracts), and certification as solutions to adverse selection, with health insurance and labour market examples. Reinforces Section 12.2.

3. Solving the Principal–Agent Problem (*Matt Clancy*) ~ 11 min

<https://www.youtube.com/watch?v=QcCEhmrU5Yg>

Explains the principal–agent setup, derives optimal effort under incentive contracts, and explains why flat salaries produce zero effort. Recommended before Level 2 moral hazard questions.

4. Principal–Agent Models and Moral Hazard (*BurkeyAcademy*) ~ 14 min

<https://www.youtube.com/watch?v=kdXqwucFNeU>

Derives the optimal incentive contract under moral hazard, explains the incentive–insurance trade-off, and discusses real-world applications. Reinforces Section 12.4.

12.8. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging.
 Notation: q = accident probability; L = loss; p = fraction of peaches; e = effort; β = performance weight; $\bar{U}$ = reservation utility. Attempt every question independently.

Level 1: Conceptual and Short-Answer

- ★ **[Definitions]** Define adverse selection and moral hazard. For each: (a) is the problem ex ante or ex post? (b) which party holds private information? (c) give one real-world market where it arises; (d) why is the outcome inefficient relative to the full-information benchmark?
- ★ **[Externality Connection]** In Akerlof's lemons model, why does the presence of low-quality cars impose a negative externality on sellers of high-quality cars? How does this externality differ from the production externalities in Week 11?
- ★ **[Pooling vs. Separating]** Explain the difference between a pooling and separating equilibrium in the used-car market. Under what conditions does each arise? Which is more likely to involve a complete market failure, and why?
- ★ **[Moral Hazard Debate]** A student argues: "Moral hazard is not really a market failure — it is just people behaving rationally given their incentives. The solution is simply to write better contracts." Critically evaluate this statement.
- ★ **[Signalling vs. Screening]** Distinguish signalling from screening as solutions to adverse selection. For each: (a) which party takes the action? (b) what makes it credible? (c) give one real-world example.
- ★ **[Lie-Detector / Camera Test]** For each scenario, state whether the problem is adverse selection or moral hazard and apply the lie-detector/camera diagnostic: (a) a used-car seller knows the car's mechanical history; (b) a fully insured driver parks in riskier locations; (c) a job applicant knows her own productivity; (d) a CEO pursues empire-building at shareholders' expense.
- ★ **[Two-Type Lemons: Basic Calculation]** Lemon: seller \$500, buyer \$700. Peach: seller \$1,800, buyer \$2,200. Buyers cannot identify quality.
 - (a) $\mathbb{E}[\text{WTP}]$ as a function of p .
 - (b) Critical fraction p^* .
 - (c) $p = 0.80$: pooling or separating? Price range?
 - (d) $p = 0.50$: equilibrium outcome?
- ★ **[Screening: Menu of Contracts]** An insurer faces two types of drivers. High-risk (H): 40% chance of an accident causing \$10,000 loss. Low-risk (L): 10% chance of the same loss. The insurer cannot observe type.

- (a) What are the actuarially fair premiums if type is observable?
- (b) Explain in plain English why the insurer cannot simply charge each type its actuarially fair premium when types are unobservable.
- (c) The insurer offers a menu: Contract A (full coverage, premium \$4,000) and Contract B (partial coverage of \$3,000, premium \$300). For a risk-neutral high-risk driver, what is the expected net benefit from each contract? Which does the high-risk driver prefer?
- (d) What does your answer to (c) tell you about whether this particular menu successfully separates types? What adjustment could the insurer make to Contract A to achieve separation?

★ **[Optimal β^* : Comparative Statics]** The optimal performance weight in the principal-agent model is $\beta^* = 1/(1 + r\sigma^2)$.

- (a) Compute β^* when $r = 0.05$ and $\sigma^2 = 100$.
- (b) Compute β^* when σ^2 doubles to 200, holding $r = 0.05$ fixed. How does the optimal contract change?
- (c) Compute β^* when r falls to 0 (risk-neutral agent), holding $\sigma^2 = 100$. What is the economic interpretation?
- (d) A biotech startup has very uncertain profits ($\sigma^2 = 500$). A regulated utility has stable profits ($\sigma^2 = 20$). Both have CEOs with $r = 0.02$. Compare β^* for each. What does this predict about how CEO compensation packages differ between these two industries?

★ **[Explain to a Friend — Level 1]** *No formulas.* Your friend says: “I don’t see how a used-car seller knowing about a mechanical defect destroys the entire market for good cars. Surely good cars still sell?” Write 3–4 plain-English sentences explaining why, in some circumstances, they don’t.

Level 2: Intermediate Problems

★★ **[Three-Type Unravelling]** Market with three types:

Type	Fraction	Seller’s min	Buyer’s max WTP
Low	0.50	\$500	\$700
Medium	0.30	\$1,500	\$1,900
High	0.20	\$3,000	\$3,600

- (a) Pooled $\mathbb{E}[\text{WTP}]$ with all three present.
- (b) Do high types stay? Why?
- (c) After high exit, recompute $\mathbb{E}[\text{WTP}]$ using updated fractions. What is the buyer’s belief at this stage? Do medium types stay?

(d) Final equilibrium.

★★ [CEO Effort Under a Linear Contract] $\pi = 20e$, $c(e) = 2e^2$, contract $w = w_0 + \beta\pi$.

- (a) Optimal effort $e^*(\beta)$.
- (b) Effort at $\beta = 0$ (flat salary)?
- (c) Effort at $\beta = 0.5$?
- (d) First-best effort e^{FB} (maximise total surplus)?
- (e) At what β is first-best achieved?

★★ [CEO Participation Constraint] Continue Q11. Reservation utility $\bar{U} = 20$; CEO risk-neutral.

- (a) CEO's utility at $e^*(\beta)$; find w_0 when PC binds.
- (b) Shareholders' expected payoff as a function of β only.
- (c) Optimal β^* .
- (d) Equilibrium effort, CEO pay, and shareholders' payoff.

★★ [Insurance Adverse Selection] Type H (high-risk): $q_H = 0.30$, $L = \$8,000$. Type L: $q_L = 0.05$, $L = \$8,000$. Equal fractions.

- (a) Actuarially fair premium for each type separately.
- (b) Pooled premium if insurer cannot distinguish types.
- (c) Do low-risk drivers participate at the pooled premium (assume risk-neutrality)?
- (d) If low-risk exit, what is the break-even premium?

★★ [Data-Based: Lemons Market across Compositions] The table tracks a used-car market (lemon WTP \$1,200 / reserve \$1,000; peach WTP \$2,400 / reserve \$2,000) as the fraction of peaches p varies.

p	$\mathbb{E}[\text{WTP}]$	Pooling?	Equilibrium price range	Welfare loss
0.90				
0.75				
0.67				
0.50				
0.25				
0.00				

- (a) Fill in all cells. Define welfare loss as the value of destroyed peach transactions.
- (b) At which value(s) of p is the equilibrium exactly on the knife-edge?

- (c) Describe how the market transitions as p falls from 0.90 to 0.00. When does the first discontinuous change occur?
 - (d) If the government introduces a mandatory car inspection programme (revealing type at cost \$100 per car), at what values of p does this intervention raise total welfare?
 - (e) Sketch the relationship between p and total welfare (under asymmetric information). Is it monotone? Why or why not?
- ★★ **[Signalling in the Labour Market — Basic]** High-ability (H): productivity \$80,000. Low-ability (L): \$40,000. Equal fractions. MBA costs: $c_H = \$15,000$; $c_L = \$35,000$.
- (a) Wage under symmetric ignorance.
 - (b) Are all workers willing to work at this wage?
 - (c) Can the MBA serve as a credible signal? Check both ICCs.
 - (d) Who gains and who loses from the separating equilibrium relative to the pooling equilibrium?
- ★★ **[Continuum Lemons Model]** Quality $q \sim U[0, 1]$; $v_s(q) = q$; $v_b(q) = 1.5q$.
- (a) Under full information: which trades occur? Total gains?
 - (b) Under asymmetric information, cars $q \leq \bar{q}$ offered. Average quality?
 - (c) Expected price buyers pay.
 - (d) Show that $\bar{q} = 0$ is the unique equilibrium.
 - (e) Welfare loss from complete unravelling?
- ★★ **[Principal-Agent: Continuous Effort]** Project succeeds with probability e ; success yields \$100; failure yields \$0. $c(e) = 50e^2$.
- (a) First-best effort.
 - (b) Under a fixed wage w , what effort does the agent choose?
 - (c) Under (w_s, w_f) , write the agent's FOC.
 - (d) If risk-neutral and $\bar{U} = 0$, what contract achieves first-best?
- ★★ **[Explain to a Friend — Level 2]** Your friend says: "I understand the CEO should get performance pay. But my boss once told me that when they tied everyone's bonus to sales numbers, service quality tanked. Isn't that backwards?" Write 4–5 sentences explaining the multi-tasking problem using the Wells Fargo example.

Level 3: Advanced Problems

★★★ **[Signalling — Full Analysis]** H -workers: \$120,000; L -workers: \$50,000. Equal fractions. Education costs $c_H(s) = s$; $c_L(s) = 2s$ (thousands).

- (a) Pooling wage.
- (b) ICC conditions for both types.
- (c) Valid range of s^* .
- (d) Is education productive in this model? Social value?
- (e) Total signalling cost; welfare under pooling vs. separating.

★★★ **[Incentive–Insurance Trade-off: Full Derivation]** $\pi = e + \varepsilon$; $\varepsilon \sim N(0, \sigma^2)$; CEO utility represented by certainty equivalent: $CE = \mathbb{E}[w] - c(e) - \frac{1}{2}r \cdot \text{Var}(w)$; $c(e) = e^2/2$; contract $w = s + b\pi$.

- (a) Write out $\mathbb{E}[w]$ and $\text{Var}(w)$ explicitly in terms of s, b, e , and σ^2 .
- (b) Write the CEO's certainty equivalent CE and derive the optimal effort $e^*(b)$ from the CEO's first-order condition. Show every step.
- (c) Impose the participation constraint $CE \geq \bar{U}$ and show that at the optimum it must bind. Solve for s as a function of b, r, σ^2 , and $\bar{U}$.
- (d) Substitute the binding PC into the principal's objective and maximise over b to show $b^* = 1/(1 + r\sigma^2)$. Show the first-order condition explicitly.
- (e) Interpret: how do r and σ^2 affect b^* ? Provide the numerical example from the text ($r = 0.02$, $\sigma^2 = 100$ then $r = 0.04$) and explain the economic intuition for each comparative static.

★★★ **[Moral Hazard: Two Effort Levels]** High effort e_H : $\Pr(\pi_H) = 0.8$, $\Pr(\pi_L) = 0.2$. Low effort e_L : $\Pr(\pi_H) = 0.3$, $\Pr(\pi_L) = 0.7$. $\pi_H = \$200$, $\pi_L = \$50$. Agent: $U = \sqrt{w} - c(e)$; $c(e_H) = 4$, $c(e_L) = 0$; $\bar{U} = 4$.

- (a) Expected profits under each effort level.
- (b) First-best wage w^{FB} if effort is observable.
- (c) PC and ICC for (w_H, w_L) under unobservable effort. State clearly what each constraint requires and why.
- (d) Solve for optimal (w_H, w_L) when both constraints bind.
- (e) Cost of asymmetric information (expected wage difference). Explain what this extra cost represents economically.

★★★ **[Adverse Selection in Credit Markets]** Bank lends \$1,000. Safe project: $p_S = 0.9$, $R_S = \$1,500$. Risky: $p_R = 0.5$, $R_R = \$2,500$. Equal fractions. Entrepreneur repays

1,000(1 + r) on success, nothing on failure.

- (a) Expected return of each project (gross of repayment).
- (b) Each type's expected profit at interest rate r .
- (c) Rate r^* at which safe types are indifferent.
- (d) For $r > r^*$, what happens? Break-even rate for risky only?
- (e) Credit-rationing implication.

★★★ [Signalling and Social Waste] H : \$100,000; L : \$40,000. Equal fractions. $c_H(s) = s$; $c_L(s) = 3s$ (thousands).

- (a) Pooling wage.
- (b) ICCs for separating equilibrium at s^* .
- (c) Valid range of s^* .
- (d) Minimum-cost separating equilibrium.
- (e) At $s^* = 20$: is direct verification at \$5,000/worker socially superior? Compare total costs.
- (f) When does signalling destroy social value relative to pooling? State the general condition.

★★★ [Multi-Tasking and Incentive Design] Teacher effort (e_T, e_W) : $T = e_T + \varepsilon_T$, $W = e_W + \varepsilon_W$. Cost: $c = \frac{1}{2}e_T^2 + \frac{1}{2}e_W^2$. Social benefit $= e_T + e_W$. Only T is contractible.

- (a) Socially optimal (e_T^*, e_W^*) .
- (b) Under $w = s + bT$: teacher's optimal (e_T, e_W) as function of b .
- (c) Show $e_W = 0$ for any $b > 0$. Interpret.
- (d) District's optimal b^* .
- (e) Total social benefit vs. first-best. Cost of distortion?
- (f) Two institutional remedies.

★★★ [AS and MH Together] Apex Insurance faces both problems simultaneously. Careful drivers: $q_C = 0.10$. Reckless: $q_R = 0.40$. Loss \$20,000. Equal fractions. Precaution costs \$1,000 but reduces accident probability by 0.10. Under full insurance, no precaution taken.

- (a) Pooled premium with full coverage and no precaution.
- (b) Do careful types participate at this premium? (Assume risk-neutrality and compare expected loss to pooled premium.)

- (c) If only reckless participate (no precaution), break-even premium.
 - (d) A co-payment of \$2,000 per accident is introduced. Will reckless types take precaution? Compare the expected saving from precaution to its cost.
 - (e) Does the co-payment help screen careful from reckless types? Explain how it changes the adverse selection problem.
 - (f) How does the co-payment address both adverse selection and moral hazard simultaneously? Identify clearly which mechanism each effect operates through. What is the limitation of a co-payment as a policy tool?
- ★★★ **[Explain to a Friend — Level 3]** Your friend, a law student, says: “The 2008 financial crisis was just greed and fraud. You economists always over-complicate things with adverse selection and moral hazard. Isn’t the real story just that bankers were dishonest?” Write a 5–6 sentence response addressing: (a) the specific role of adverse selection in the securitisation market; (b) the specific moral hazard created by implicit government guarantees; (c) why these are market failures even without dishonesty; (d) what the correct policy response is according to the theory; and (e) one limitation of the informational explanation.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.